

Alvin Litani Liauw

Machine learning engineer

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Machine learning engineer focused on building and deploying ML systems: time-series forecasting for wind generation served through a FastAPI service on Google Cloud Run, retrieval-augmented generation over enterprise documentation, and computer vision similarity search. Returned to technical work in 2025 after moving to other fields, and recently finishing a graduate certificate in AI software development. Earlier background has nearly three years of production infrastructure experience maintaining Java and Ruby applications, and over four years in client-facing technical roles.

Eligible to work in Canada without sponsorship. Open to relocation anywhere in Canada, or remote.

Selected projects

Wind power forecasting, Ontario

XGBoost, FastAPI, Google Cloud Run - github.com/alvinlitani/wind-power-forecast | live API

Day-ahead hourly forecasts for all 45 IESO-reporting wind plants in Ontario (4,943 MW, ~90% of provincial wind capacity). Per-site XGBoost models predict capacity factor and are trained on historical weather forecasts rather than actuals to avoid train/serve data mismatch. 11.3% capacity-weighted nMAE over a full-year 2025 test set (394,006 site-hours); 0.37 MAE skill score against a persistence baseline in live daily runs. Prefect Cloud schedules a Cloud Run Job writing to GCS, with a separate FastAPI service serving predictions and fail-closed data quality gates throughout.

Celebrity face matching

facenet-pytorch, FAISS, Gradio - github.com/alvinlitani/Celebrity-Face-Matching | live demo

A computer vision app that takes a personal photo as input and finds which celebrity in CelebA dataset that the person most resemble using face embeddings and nearest-neighbor search. It is built with facenet-pytorch and FAISS. Hosted on HuggingFace.

Product manual assistant

LangChain, ChromaDB, Hugging Face - github.com/alvinlitani/RAG-for-Product-Manual-Assistant | live demo

A RAG (Retrieval-Augmented Generation) chatbot that answers technical questions about product documentation. Built with LangChain v1, HuggingFace, and ChromaDB. Hosted on HuggingFace.

Experience

Machine Learning Engineer

Independent, Ottawa, Ontario (2025–present)

Self-directed development and deployment of production ML systems.

- Sole engineer on three machine learning systems bringing them from design all the way to deployment. All three are live and open to the public.
- The wind power forecasting project runs daily with no missed executions to date. Built monitoring and data quality safeguards that exit the run rather than publish predictions when the input data looks wrong.
- Sourced and reconciled four public datasets that shared no common ID (grid operator records, a national turbine database, provincial filings, and weather data).

Warehouse supervisor

PT Bangun Prima Sukses, Indonesia (2019-2023)

- Supervised a team of 10+ and a fleet of 8 delivery trucks covering scheduling, training, and day-to-day performance management.
- Coordinated delivery schedules to 20+ active construction sites.

Technical manager

PT Perkasa Pilar Utama, Indonesia (2015-2019)

IT consultancy building systems for government and business clients. Introducing new Internet of Things/IoT (LoRaWAN) technology to the Indonesian market. Continue to advise informally on their adoption of agentic coding tools (2025-

present).

- Negotiated partnership and joint-venture agreements including contract terms with global technology vendors across 4 countries.
- Ran multi-day LoRaWAN trials at 5 client sites (cellular base stations, a public water pipeline, a plantation, a frozen food warehouse, and a mining site) handling logistics and leading the on-site technical demonstrations.
- Evaluated 20+ vendor hardware and software platforms for product-market fit against enterprise requirements in the Indonesian market.
- Acted as technical point of contact across 4 enterprise accounts and coordinated 5-person teams spanning engineering, logistics, and sales.
- Created technical pitch documents and supporting data analysis for business proposals to clients.

Application administrator

PT Midtrans, Indonesia (2013-2015)

Payment gateway processing card and bank transfers for hundreds of Indonesian e-commerce merchants. Part of the infrastructure team and 2 DevOps teams.

- Monitored and tuned 5 Java and Ruby applications across 3 Linux servers, maintaining 99.9% uptime on live payment processing.
- Introduced application performance telemetry with New Relic, cutting incident detection from hours to roughly 15 minutes.
- Traced production failures through application and system logs to isolate root causes.
- Deployed releases and ran rollbacks weekly, handling around 2 production incidents a week on a 2-person on-call rotation.

Technical sales and merchant support

PT Midtrans, Indonesia (2012-2013)

Early-stage startup with no dedicated technical support for the sales team.

- Built and modified 3 PHP payment extensions for the Magento, Drupal, and WooCommerce e-commerce platforms.
- Supported around 2 merchant integrations a week, working with client engineering teams to troubleshoot API and transaction failures.
- Joined near-daily client meetings as the technical contact alongside sales, covering roughly 4 prospects a week.

Education

Graduate Certificate in Artificial Intelligence Software Development

Algonquin College, Ontario (2024-2025)

Graduate Certificate in Supply Chain Management

Algonquin College, Ontario (2023-2024)

Bachelor of Computer Science and Technology

University of Sydney, Australia (2008-2011)

Skills

Languages - Python, SQL; Java, Ruby, PHP (working exposure)

Machine learning and AI - PyTorch, facenet-pytorch, scikit-learn, XGBoost, LangChain, ChromaDB, FAISS, Hugging Face, Pandas, NumPy

Tools and infrastructure - Linux, Docker, Git, FastAPI, Google Cloud Run, Hugging Face Spaces, Gradio, New Relic, Prefect, Weights & Biases, Google Cloud Storage

Concepts - Time-series forecasting, RAG, vector databases, embeddings, LLM APIs, prompt engineering, NLP, sentiment analysis, computer vision

Spoken languages - English (professional), Indonesian (native)